

Lecture Scribe
RANDOM VARIABLES, DISTRIBUTIONS &
EXPECTATION

March 20, 2026

1 (1) Previous Lecture Recap: Random Variables (RVs)

1.1 1.1 Definition of a Random Variable

Let:

SSS = Sample Space of a random experiment

$\omega \in S$ = an outcome

A Random Variable (RV) is defined as a function:

$$X : S \rightarrow \mathbb{R}$$

This means:

For every outcome $\omega \in S$, the function assigns a real number.

So,

$$X(\omega) = \text{real number}$$

Thus, a random variable converts outcomes of an experiment into numerical values.

1.2 1.2 Types of Random Variables

1.2.1 (A) Discrete Random Variable

A random variable that takes countable values.

Representation:

$$X = x_1, x_2, x_3, \dots$$

1.2.2 (B) Continuous Random Variable

A random variable that takes values over a continuous interval.

Representation:

$$a \leq X \leq b$$

1.3 1.3 Probability Mass Function (PMF)

If X is discrete:

$$P(X = x_i) = p_i$$

Properties

$$0 \leq p_i \leq 1$$

$$\sum p_i = 1$$

These conditions ensure that probabilities are valid and total probability equals 1.

2 (2) Independent Events

2.1 2.1 Definition

Two events A and B are independent if:

$$P(A \cap B) = P(A)P(B)$$

Logical meaning:

The occurrence of one event does not change the probability of the other event.

2.2 2.2 Example: Two Coin Tosses

Step 1: Write Sample Space

$$S = \{HH, HT, TH, TT\}$$

Total outcomes = 4.

Step 2: Define Events

Let:

A = First toss is Head

Outcomes satisfying A :

$$\{HH, HT\}$$

Thus,

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

Let:

B = Second toss is Head

Outcomes satisfying B :

$$\{HH, TH\}$$

Thus,

$$P(B) = \frac{2}{4} = \frac{1}{2}$$

Step 3: Find Intersection

$$A \cap B = \{HH\}$$

Thus,

$$P(A \cap B) = \frac{1}{4}$$

Step 4: Verify Independence Condition

$$P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Since:

$$P(A \cap B) = P(A)P(B)$$

Therefore, A and B are independent.

3 (3) Types of Discrete Random Variables

3.1 3.1 Bernoulli Random Variable

3.1.1 Definition

A random variable with two possible outcomes:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

3.1.2 PMF

$$P(X = x) = p^x(1 - p)^{1-x}$$

where $x = 0, 1$

3.1.3 Expectation (Step-by-Step)

By definition:

$$E[X] = \sum xP(X = x)$$

Substitute possible values:

$$E[X] = 1 \cdot p + 0 \cdot (1 - p)$$

Compute terms:

$$1 \cdot p = p \quad 0 \cdot (1 - p) = 0$$

Add:

$$E[X] = p + 0 = p$$

3.2 3.2 Binomial Random Variable

3.2.1 Definition

If:

n independent trials

Each trial has probability of success p

Then:

X = Number of successes

3.2.2 PMF

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

3.2.3 Expectation (Step-by-Step Logical Derivation)

Let:

$$X = X_1 + X_2 + \cdots + X_n$$

Each X_i is Bernoulli.

We know:

$$E[X_i] = p$$

Now apply linearity:

$E[X] = E[X_1 + X_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n]$
 Substitute:
 $= p + p + \cdots + p$
 Since there are n terms:
 $E[X] = np$

3.3 Geometric Random Variable

3.3.1 Definition

Number of trials required to get the first success.

3.3.2 PMF

$$P(X = k) = (1 - p)^{k-1}p$$

3.3.3 Expectation (Step-by-Step Derivation)

$$E[X] = \sum_{k=1}^{\infty} k(1 - p)^{k-1}p$$

Factor p :

$$= p \sum_{k=1}^{\infty} k(1 - p)^{k-1}$$

Let:

$$r = 1 - p$$

We use the known series result:

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}$$

Substitute:

$$1 - r = p$$

Thus:

$$\sum k(1 - p)^{k-1} = \frac{1}{p^2}$$

Therefore:

$$E[X] = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

3.4 Poisson Random Variable

3.4.1 Definition

Used when events occur independently in fixed interval.

Parameter:

$$\lambda > 0$$

3.4.2 PMF

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

3.4.3 Expectation (Step-by-Step)

$$E[X] = \sum_{k=0}^{\infty} k \frac{e^{-\lambda} \lambda^k}{k!}$$

Rewrite:

$\frac{k}{k!} = \frac{1}{(k-1)!}$
 Thus:
 $E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$
 Let $m = k - 1$:
 $= \lambda e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!}$
 Recognize exponential expansion:
 $\sum_{m=0}^{\infty} \frac{\lambda^m}{m!} = e^{\lambda}$
 Thus:
 $E[X] = \lambda e^{-\lambda} e^{\lambda} = \lambda$

4 (4) Expectation of Random Variables

4.1 4.1 Definition

$$E[X] = \sum xP(X = x)$$

4.2 4.2 Expectation of Function

$$E[g(X)] = \sum g(x)P(X = x)$$

4.2.1 Example

Given:

$$P(X = 1) = 0.5, P(X = 2) = 0.5$$

Find:

$$E[X^2] = 1^2(0.5) + 2^2(0.5) = 1(0.5) + 4(0.5) = 0.5 + 2 = 2.5$$

4.3 4.3 Linearity of Expectation

$$E[aX + b] = \sum (ax + b)P(X = x) = a \sum xP(X = x) + b \sum P(X = x) = aE[X] + b(1) = aE[X] + b$$

4.4 4.4 Variance Derivation

$$\text{Var}(X) = E[(X - \mu)^2]$$

Expand:

$$(X - \mu)^2 = X^2 - 2\mu X + \mu^2$$

Take expectation:

$$E[X^2] - 2\mu E[X] + \mu^2$$

Since $E[X] = \mu$:

$$= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$$

Thus:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

4.5 4.5 n Moments

$$\mu_n = E[X^n]$$

Special cases:

2nd central moment $\rightarrow$ Variance

3rd $\rightarrow$ Skewness

4th $\rightarrow$ Kurtosis

5 (5) Cumulative Distribution Function (CDF)

5.1 Definition

$$F(x) = P(X \leq x)$$

5.2 Properties

$$0 \leq F(x) \leq 1$$

Non-decreasing

Right continuous

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow \infty} F(x) = 1$$

5.3 Example

Given:

$$P(X = 1) = 0.4, P(X = 2) = 0.6$$

Then:

For $x < 1$:

$$F(x) = 0$$

For $1 \leq x < 2$:

$$F(x) = 0.4$$

For $x \geq 2$:

$$F(x) = 1$$

6 (6) Probability Density Function (PDF)

6.1 Definition

$$f(x) \geq 0 \quad \int_{-\infty}^{\infty} f(x)dx = 1$$

6.2 Probability Between Two Points

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

6.3 Relation Between CDF and PDF

$$F(x) = \int_{-\infty}^x f(t)dt$$

Differentiate:

$$f(x) = \frac{d}{dx}F(x)$$

7 FINAL REVISION SUMMARY

- RV = Function from sample space to real numbers
- Independence = $P(A \cap B) = P(A)P(B)$
- Bernoulli $\rightarrow E[X] = p$
- Binomial $\rightarrow E[X] = np$
- Geometric $\rightarrow E[X] = 1/p$
- Poisson $\rightarrow E[X] = \lambda$
- Variance $\rightarrow E[X^2] - (E[X])^2$
- CDF $\rightarrow F(x) = P(X \leq x)$
- PDF integrates to 1